

Experiment 3-High Wind: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1827	6.853	5.221	14.850	5.6%	21.7%	35.1%	19.576	46.210	13.460	43.512	0.685	1.434	1.543
Proposed	1763	5.552	3.044	14.521	14.3%	36.7%	49.7%	25.460	76.820	11.703	31.991	0.739	1.554	1.671
PPO-GRU	1737	5.829	3.038	15.868	14.2%	33.8%	49.5%	25.612	87.483	11.224	25.412	0.663	1.503	1.677
MPC	420	7.172	2.701	24.170	10.2%	30.0%	59.5%	2.622	4.239	1.919	3.643	0.153	0.171	0.223

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1827	91.3%	173	8.7%	8.3%	166	96.0%	7	0
Proposed	2000	1763	88.2%	237	11.8%	11.2%	221	93.2%	16	0
PPO-GRU	2000	1737	86.8%	263	13.2%	11.7%	232	88.2%	31	0
MPC	2000	420	21.0%	1580	79.0%	79.0%	1580	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	1580	1434	90.8%	1007	70.2%	7.236
Proposed	1580	1349	85.4%	732	54.3%	5.912
PPO-GRU	1580	1317	83.4%	690	52.4%	6.109